

# Economic History II

Bureaucracy and public service delivery

FGV EPGE

# Outline of the class

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- **Two papers**, taken separately. All on bureaucrats and what the state delivers.
- Aneja and Xu (2024): civil service reform in the US post office during the Gilded Age. *American Economic Review*.
- Akerman, Moscona, Pellegrina and Sastry (2025): Embrapa and Brazil's agricultural revolution. Working paper.

# Post offices, patronage and reform

Aneja and Xu (2024), *Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age*, AER

# The question

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- An impartial civil service with lifelong careers: the standard prescription for a working state, since Weber.
- Direct evidence that such reforms improve state effectiveness: scarce.
- The setting is the **Pendleton Act of 1883**, the first nationwide civil service reform in the United States.
- The organization is the post office, the largest civilian agency of the federal government.

# Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age<sup>†</sup>

By ABHAY ANEJA AND GUO XU\*

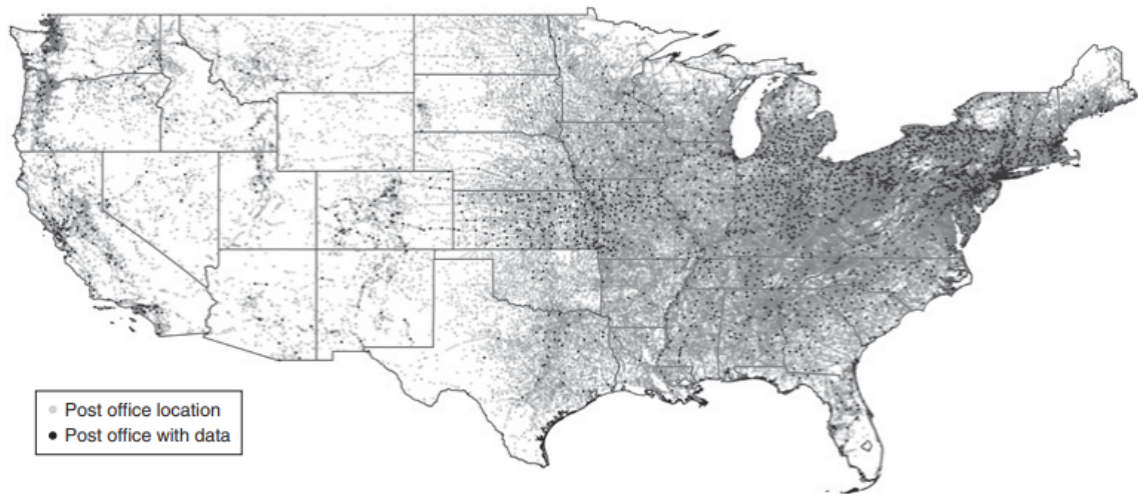
*We use newly digitized records from the post office to study the effects of strengthened state capacity between 1875 and 1901. Exploiting the implementation of the Pendleton Act—a landmark statute that shielded bureaucrats from political interference—across US cities over two waves, we find that civil service reform reduced postal delivery errors and increased productivity. These improvements were most pronounced during election years when the reform dampened bureaucratic turnover. We provide suggestive evidence that reformed cities witnessed declining local partisan newspapers. Separating politics from administration, therefore, not only improved state effectiveness but also weakened the role of local politics. (JEL D24, D73, H83, J45, L82, L87, N41)*

## The post office in the nineteenth century

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- At the onset of the Pendleton Act, postal workers were **34 percent of the entire federal workforce**.
- Principal method of long-distance communication in an expanding nation.
- Offices in virtually all parts of the country, all performing the same function: collection and delivery of mail, by clerks and carriers.
- That uniformity is what makes performance comparable across cities and years.
- John (1995): for most Americans, “the postal system was the central government”.

Panel A. All post offices



# The spoils system

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- Before 1883: patronage-based hiring and firing, with elected politicians holding discretion over appointments.
- Patronage employees campaigned for their benefactors and paid part of their salary back as **political assessments**.
- Eaton (1885) on what the parties extracted: “the plunder of the salaries of the public servants for filling the party treasury”.
- Careers tied to particular politicians, so turnover was high.
- One report on the New York City Post Office: “incompetency, neglect, confusion and drunkenness”.



# The Pendleton Act (1883)

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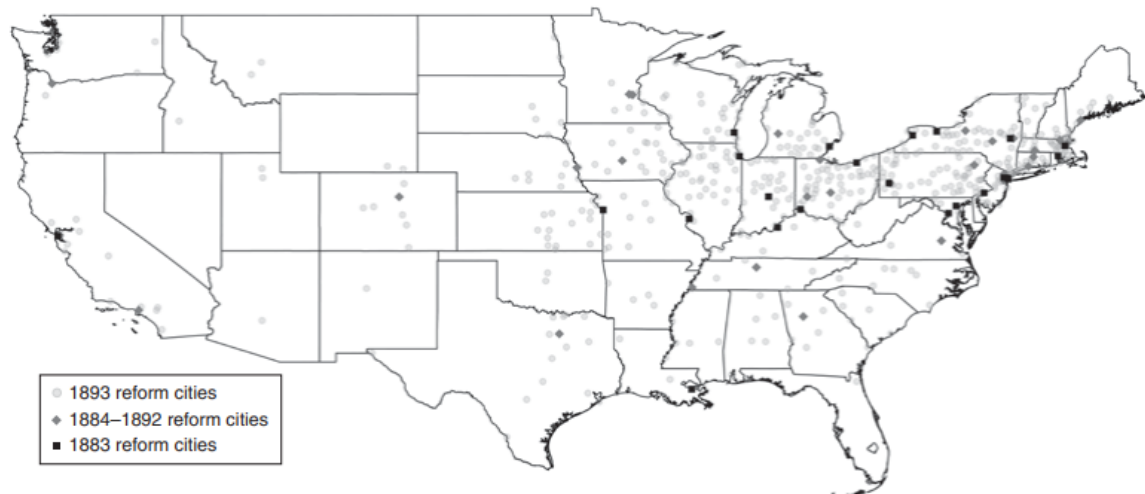
- Discretionary appointments replaced by rule-based hiring through open, competitive **examinations**.
- Classified workers protected from removal without cause.
- Assessments outlawed for the whole federal government.
- Catalyzed by the assassination of President Garfield in 1881 by a disappointed office seeker.
- About 10 percent of federal employees were classified at the outset.

## The reform arrived in waves

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- Coverage was decided in Washington and keyed to size, not to local preference.
- 1883: post offices with at least 50 employees. **23 cities.**
- 1893: Harrison's executive order classified every post office with free delivery service.  
556 more cities.
- In between, 30 cities crossed the 50-employee threshold on their own.
- By 1921, 80 percent of the federal civilian workforce was covered.

Panel B. Expansion of civil service reform



# The data

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- Large-scale digitization of the post office's own annual reports.
- **Delivery errors**: errors on returned slips, city by year, 1879 to 1901, 2,319 cities.
- Free delivery statistics: volume per carrier and cost per volume, 512 large cities, 1875 to 1891.
- Official Registers: every clerk and carrier, 1877 to 1901, 297,932 individual-year observations, linked to the full-count censuses of 1880 and 1900 (36 percent match).
- Newspaper directories: partisan and independent papers by city and election year.

## C Census linking

We match the personnel records from the Official Registers of the United States (the “Registers”) to the U.S. full count Decennial Census to obtain additional individual background characteristics. Each postal worker is matched based on the full name (first name, middle name/initial, last name), birth state, and current state of employment. We proceed by using different combinations, successively relaxing the matching restrictions:

- Step 1: First name + middle name + last name + birth state + current state
- Step 2: First name + middle name initial + last name + birth state + current state
- Step 3: First name + last name + birth state + current state
- Step 4: First name + last name + birth state

Given the limited number of variables that are available for matching, we pursue a conservative approach to ensure we do not overmatch by linking incorrect individuals (i.e., false positives). To start with, we always match individual names exactly. Second, we discard candidate matches of Census respondents who were younger than

Table C1: Characteristics of Census-linked vs. non-linked workers

|                                            | (1)                         | (2)       | (3)                | (4)               |
|--------------------------------------------|-----------------------------|-----------|--------------------|-------------------|
|                                            | <u>Mean characteristics</u> |           | <u>Differences</u> |                   |
|                                            | Matched                     | Unmatched | Raw                | Conditional       |
| <i>Panel A: Individual characteristics</i> |                             |           |                    |                   |
| Log(Salary)                                | 5.80                        | 5.83      | -0.025<br>(0.007)  | 0.016<br>(0.007)  |
| Clerk                                      | 0.59                        | 0.60      | -0.011<br>(0.002)  | -0.006<br>(0.002) |
| Same state                                 | 0.52                        | 0.66      | -0.141<br>(0.002)  | -0.147<br>(0.002) |
| Foreign-born                               | 0.11                        | 0.10      | 0.008<br>(0.001)   | 0.011<br>(0.002)  |
| German                                     | 0.02                        | 0.02      | 0.000<br>(0.000)   | 0.002<br>(0.001)  |
| Irish                                      | 0.022                       | 0.021     | 0.001<br>(0.001)   | 0.002<br>(0.001)  |

## Where the error counts come from

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- Clerks bundled mail by destination and attached a **facing slip** naming the destination and the sending clerk.
- Railway Mail Service clerks sorted the mail in transit and recorded every misdirected item.
- Errors were double-checked by a second examiner and reported back to supervisors.
- The Railway Mail Service was an independent arm of the post office, outside the city-level rollout.

## How the effect is measured

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- Compare cities **before and after** the reform reaches them with cities not yet reformed.
- The rollout rule was federal and size-based, so the timing was not a local political choice.
- Errors were recorded by a body with no stake in the local office, which limits strategic reporting.
- Effects traced year by year around the arrival of the reform, and separately for each wave.



# Results

## Fewer delivery errors

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- Reformed cities: delivery errors fall by **19 to 22 percent**.
- Holds controlling for the size rule that assigned treatment, and for a wide set of baseline city characteristics.
- No difference in trend beforehand. The gap opens at the reform, widens, then levels out.
- The same dynamic pattern in 1883 and in 1893, a decade apart.
- Not driven by any single city or state.

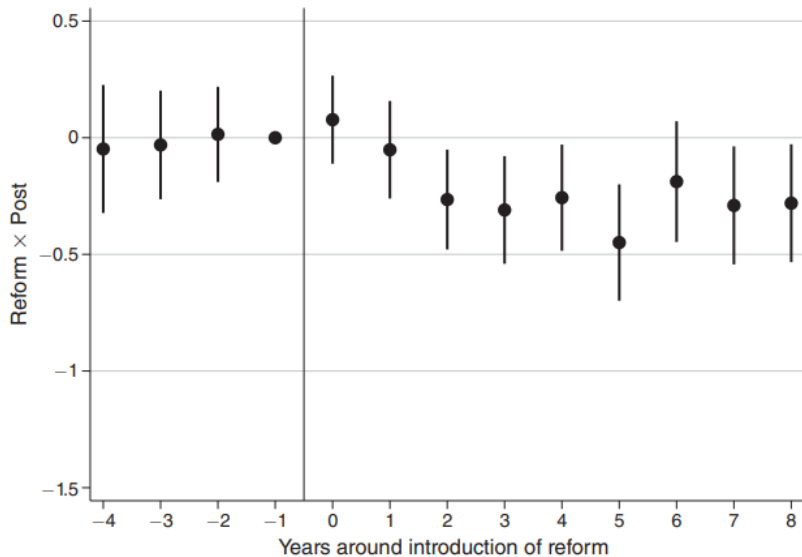
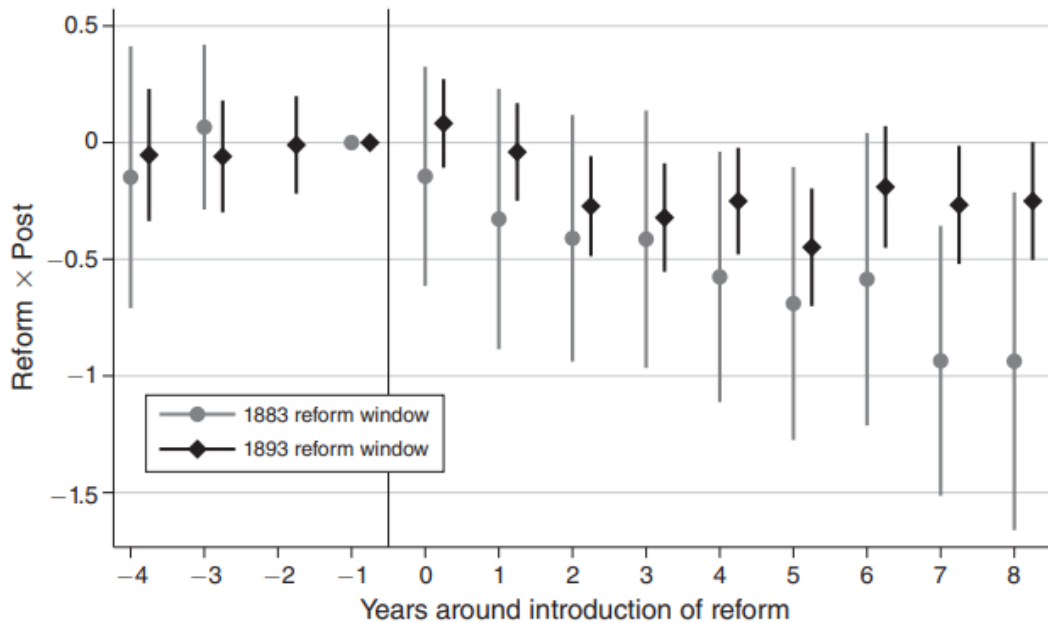


FIGURE 2. DELIVERY ERRORS FOR REFORM VERSUS NONREFORM CITIES AROUND REFORM YEARS

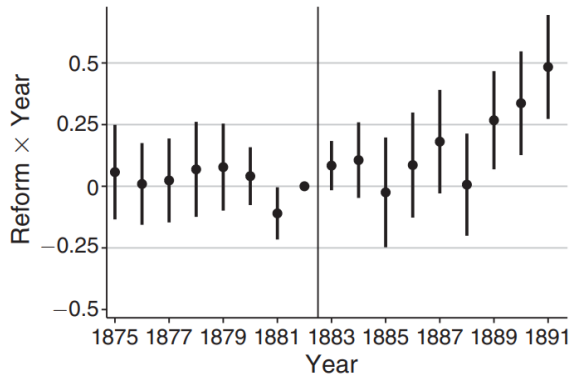


## Higher productivity

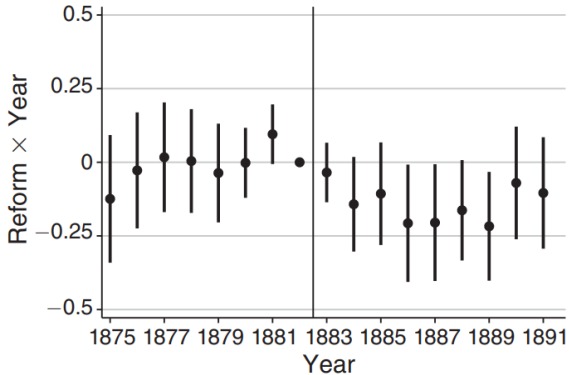
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- A second, independent record series: the City Free Delivery statistics, for the 1883 wave.
- Mail volume per carrier rises **8 to 14 percent**.
- Cost per volume falls 10 to 13 percent.
- Two measures from two sources, pointing the same way.

Panel A. Mail per carrier



Panel B. Cost per volume



## Ruling out the simple explanations

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- More staff? Total postal employment moves 1.4 to 1.7 percent, indistinguishable from zero. Errors per worker fall as well.
- Different recruits? Home state, foreign birth, nationality, age, literacy, race and pay of new hires are all unchanged.
- Less work? Mail volume, if anything, rose.
- What changes is the **link between volume and errors**.

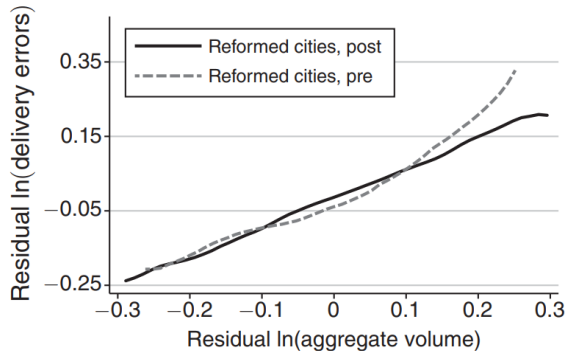
## Volume and errors

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- Before the reform, delivery errors rose with the volume of mail handled.
- After the reform, that relationship flattens in reformed cities.
- No comparable change among unreformed cities over the same years.
- The reform altered the “technology” of production rather than the workload.



Panel A. Reformed cities



Panel B. Unreformed cities

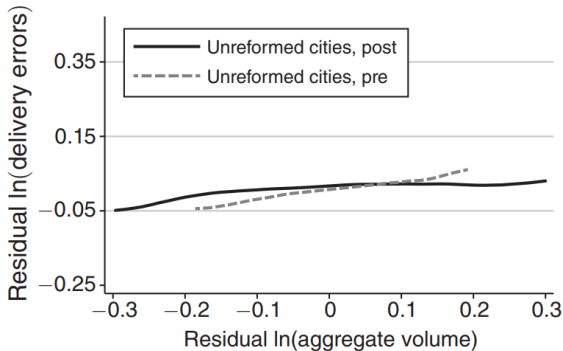
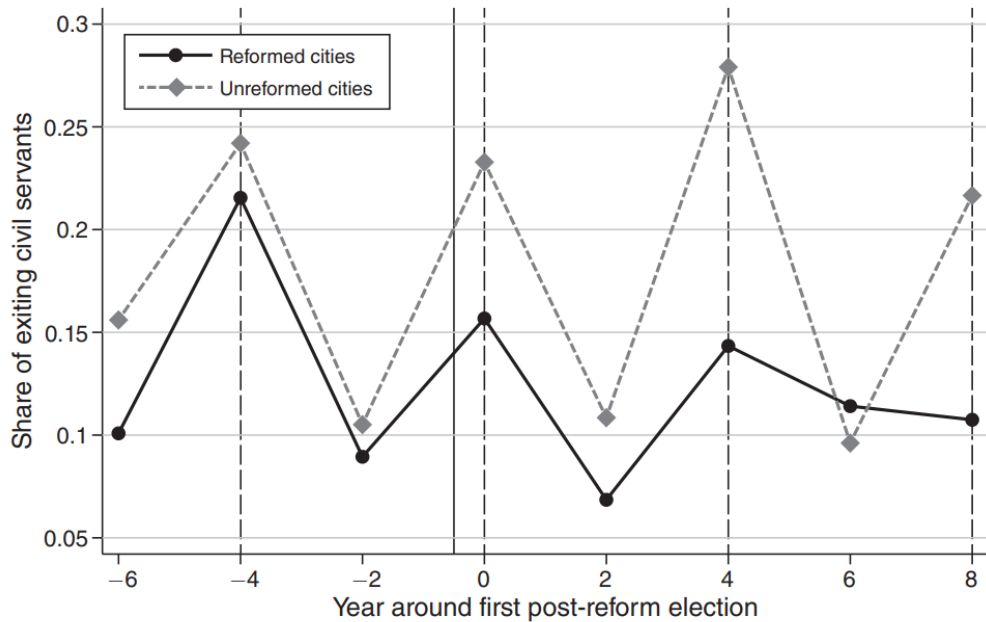


FIGURE 5. DELIVERY ERRORS AND AGGREGATE MAIL VOLUME AROUND THE REFORM

## Turnover and the electoral cycle

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- Serving at the pleasure of politicians meant insecure tenure. Hoogenboom (1959): “morale was low in a civil service largely composed of misfits employed on a temporary basis”.
- Annual exit rate among postal workers: **17 percent**, spiking in the years after presidential elections.
- The reform cuts the exit rate by about 4 points, a relative decline of 21 to 23 percent.
- Mean years of service rise by 0.8 to 1.0, from a base of 1.7.
- The electoral spike in exits flattens once a city is reformed.



## The gains land in election years

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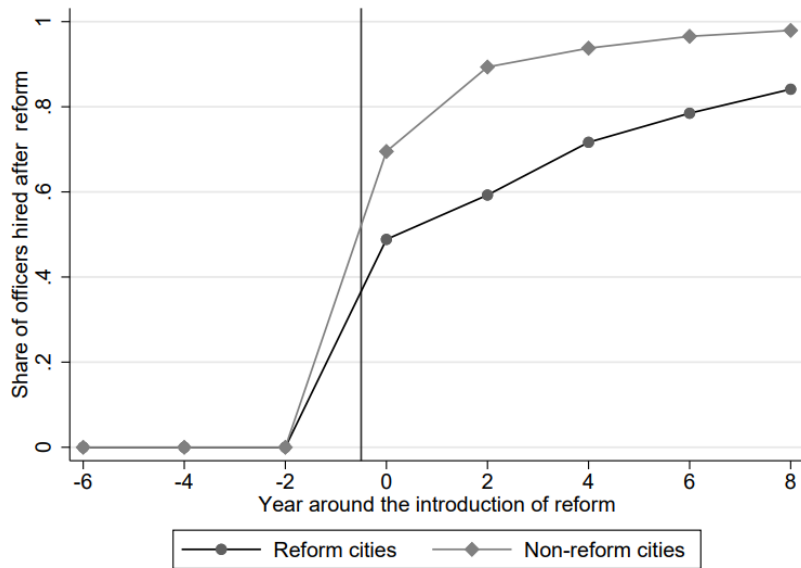
- The fall in delivery errors is driven by **election years**.
- So are the gains in volume per carrier and in cost per volume.
- Those were the years when patronage turnover previously destroyed accumulated experience.
- Retention improves for everyone serving in a reformed office, whatever their year of hire.

## Selection was too slow to be the story

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- Competitive exams applied only to new entrants, so replacing the existing staff was a protracted process.
- Eight years after the rules took effect, close to **20 percent** of the average reformed office were still patronage appointees.
- Lower exit is not confined to the post-reform entry cohort: it holds for those appointed under patronage too.
- The operating margin was the limit on arbitrary firing.

Figure A4: Share of civil servants hired after the reform event, by reform status

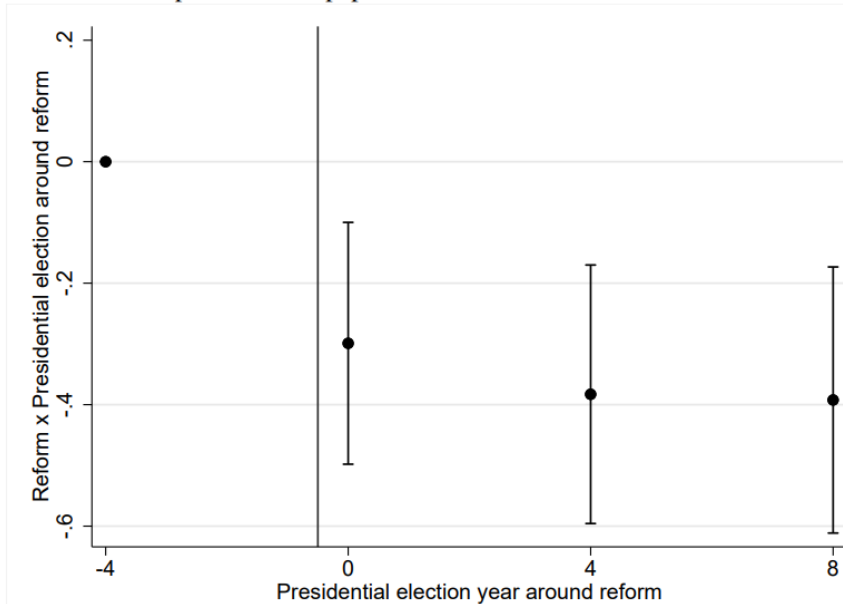


## A side effect: weaker local parties

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- Appointments were formally federal, in practice delegated to local politicians, and a source of political rents.
- Fewer patronage jobs meant less to trade for local political and financial support.
- Proxy for the strength of local parties: explicitly **partisan newspapers**, whose editors were often party officers.
- Partisan papers per city fall by 0.2 against a mean of 1.8, an 11 percent decline. Independent papers rise.
- The partisan share of circulation falls by about 4 points.

Figure A10: Number of political newspapers in reform vs. non-reform cities around reform year





## In short: Aneja and Xu

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- The Pendleton Act improved delivery and productivity in the largest civilian agency of the federal state.
- The channel was severing the tie between politicians and rank and file bureaucrats.
- Stable careers rather than a different pool of recruits.
- The authors call it a “double dividend”: a more effective bureaucracy, and weaker local parties.

# Public R&D and Brazilian agriculture

Akerman, Moscona, Pellegrina and Sastry (2025), *Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution*

# The question

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- Standard view: innovation happens in a few frontier countries, and everyone else grows by **adopting** foreign technology.
- But frontier technology in agriculture and medicine is tailored to the environments of high-income countries.
- So public R&D in locally appropriate technology may be a way out of the technology mismatch trap.
- The case is Embrapa, the public research corporation created in Brazil in 1973.

# Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution

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Notre Dame

Karthik A. Sastry§  
Princeton

November 5, 2025

## Abstract

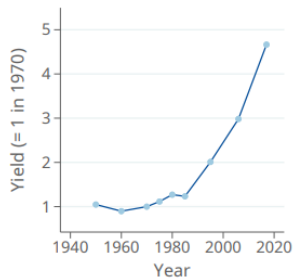
Can public R&D investment in developing countries drive productivity growth? We study this question in the context of Brazil's agricultural revolution and the Empresa Brasileira de Pesquisa Agropecuária (Embrapa), a public research corporation established in 1973 to develop locally relevant science and technology. First, using researcher-level data compiled from the resumes of all agricultural scientists in Brazil, we show that Embrapa redirected research toward locally important staple crops and toward Brazil's particular ecological conditions, in part by sustaining productive research even in remote and research-scarce regions. Second, using municipality-level panel data from nine rounds of Brazil's agricultural census, we study how Embrapa affected agricultural production. We develop an identification strategy that exploits the staggered establishment of Embrapa's research laboratories and heterogeneous exposure to the benefits from new innovation in ecologically distinct parts of the country. We find that municipality-level exposure to Embrapa significantly increased agricultural yields, driven both by expanded input use and higher residual productivity. Consistent with effects being driven by locally appropriate innovation, exposure to Embrapa predicts larger take-up of Embrapa's new crop varieties and pronounced productivity gains only for the staple crops that were the focus of Embrapa's research. Third, combining our estimates with a model, we find that public R&D investment increased national agricultural productivity by 110% with a benefit-cost ratio of 17. Embrapa's decentralized structure, in which research labs were spread across many ecological zones instead of in a single hub, explains more than half of these gains.

## Brazilian agriculture before the 1970s

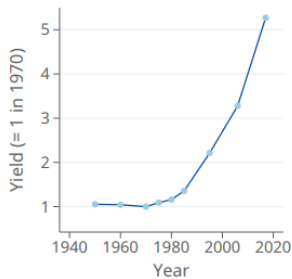
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- Yields for maize, wheat and rice close to flat between 1950 and 1970.
- Already a net food importer, increasingly reliant on foreign donations and **food aid**.
- The Green Revolution did not reach Brazil: new varieties suited neither the acidic soils nor largely rainfed farming.
- Domestic research was concentrated in wealthier states and in cash crops. By 1960 more than half the non-fruit varieties from the Campinas institute were coffee, sugarcane and cotton.

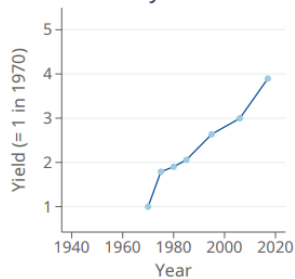
### Maize



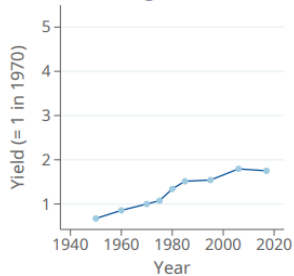
### Rice



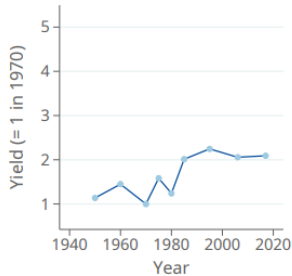
### Soybeans



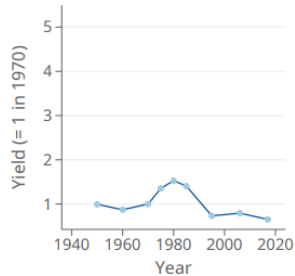
### Sugarcane



### Coffee



### Cocoa



## Embrapa: origins and design

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- Eliseu Alves, one of its founders: “The potential existed, but there was no science capable of generating technology suited to what we needed”.
- The design group named two bottlenecks: the **geographic centralization** of research, and the lack of trained personnel and careers.
- Three organizational principles: scale, geographic scope, and the legal form of a public corporation.
- Centers opened in regions with no research tradition. Priority regions: the Northeast, the Cerrado and the Amazon.

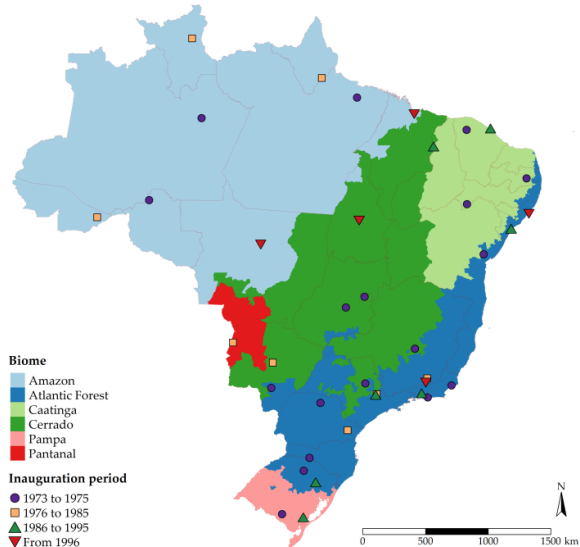
# Scale

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- By the 2000s, Embrapa's spending as a share of agricultural GDP was comparable to all public agricultural R&D in the United States.
- Budget in 2010: 1.9 billion reais, about 1.15 billion dollars. The USDA research budget that year was 2.3 billion dollars.
- About **2,300 agricultural scientists** in 2010, 40 percent of all those in the country.
- Stated goal: to study every region and ecosystem of Brazil.



**Figure 2: Embrapa's Research Centers**



## The Cerrado and soy

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- The Cerrado: two million square kilometers of tropical savanna, very low soil pH, high aluminum saturation.
- Borlaug on the view in the global research community: “nobody thought these soils were ever going to be productive”.
- Embrapa Cerrados, 1975: soil chemistry and **liming**, plus bacterial nitrogen fixation. Two Embrapa researchers later received the World Food Prize for these strands.
- Embrapa Soja, also 1975, set out to tropicalize the soybean. A first variety adapted to tropical latitudes followed in 1980.
- By the 2006 census the Cerrado supplied 42 percent of national soy, against no more than 5 percent of any of these crops in the 1970s.

# The data

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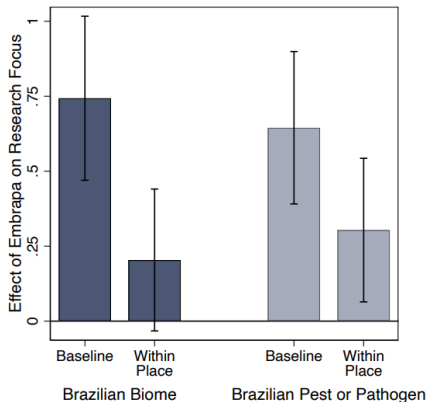
- Founding year, location and annual budget of every Embrapa center, obtained by transparency request.
- The Lattes CV platform: 35,602 agricultural researchers and about 1.3 million articles, with titles coded by crop, biome and pest.
- Nine rounds of the Agricultural Census, 1960 to 2017, digitized from scans for the early years.
- Nine **agro-climatic characteristics** per municipality: temperature, precipitation, elevation, ruggedness, growing season, soil acidity and texture.

## Embrapa redirected research

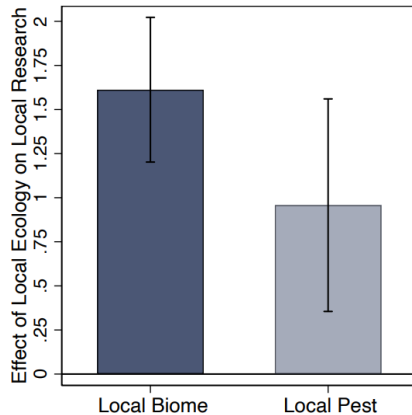
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- Embrapa articles mention a Brazilian biome **74 percent** more often than the sample average, and a Brazilian pest or pathogen 64 percent more.
- Much of that is about where the research is done: the gap shrinks sharply once location is held fixed.
- Researchers study the biome they sit in and the pests present in their state.
- Embrapa also shifted attention to staple crops (beans, cassava, maize, rice, soy, wheat), and that shift survives holding location and the researcher fixed.

**Figure 3: The Direction of Research Across Ecological Conditions**

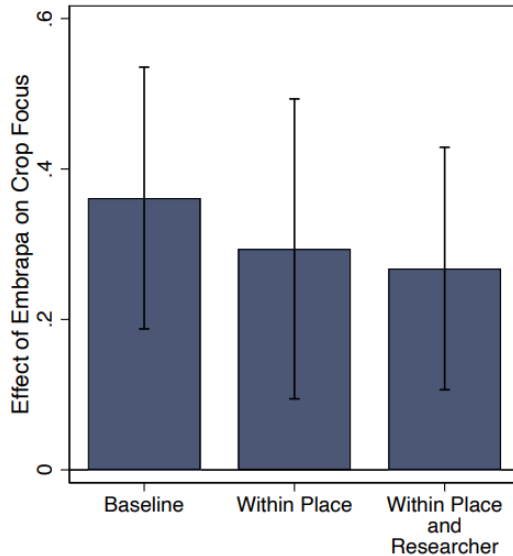


**(a)** Embrapa's Effect on Topic Focus ( $p \times i \times t$ )



**(b)** Local Research Focus ( $i \times k$ )

**Figure 4: The Direction of Research Across Crops**

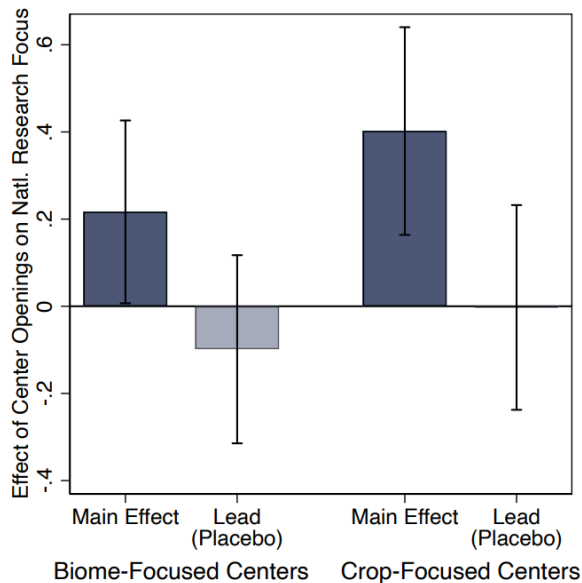


## No crowding out

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- Embrapa might simply have absorbed research that would have happened anyway.
- Opening a center focused on a crop or a biome raised **national** research on that topic.
- Part of the increase comes from researchers not employed by Embrapa.
- No movement in the year before an opening. If anything there was crowd in.

**Figure 5: Embrapa and the National Direction of Research**





## Productive research far from the hubs

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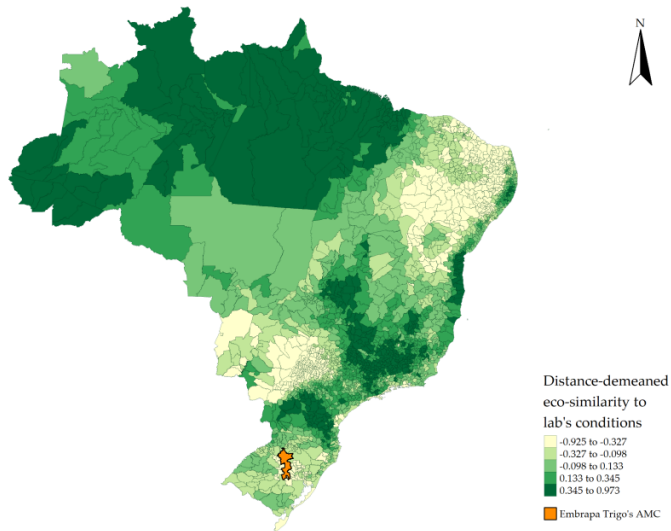
- Centers went to places with little pre-existing research capacity and no agglomeration of scientists.
- Researchers move across both employers and locations, so the two effects can be separated.
- Employment at Embrapa raised research output, and by most in the **remote regions**.
- Enough to more than offset the penalty of working away from the established hubs.
- The same holds restricting to high-impact international journals.

## Measuring exposure to Embrapa

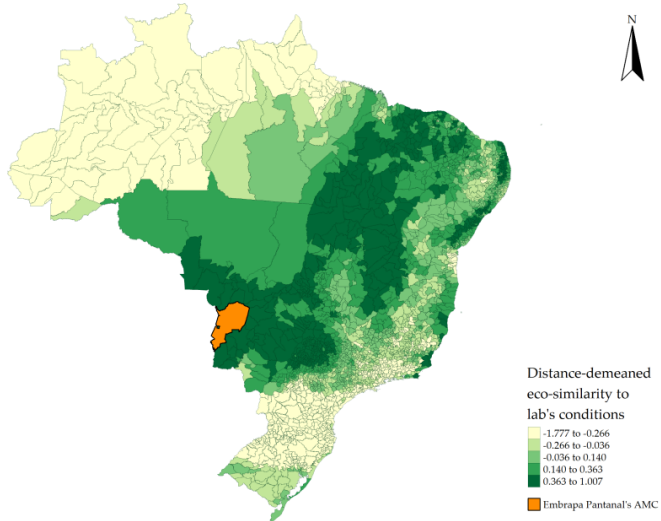
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- Technology developed for one ecology travels to similar ecologies, not to distant ones.
- **Ecological similarity** between each municipality and each center, built from the nine agro-climatic characteristics.
- Exposure of a municipality in a given year: the highest similarity among centers open by that year.
- Exposure rises as Embrapa spreads into new ecological zones.

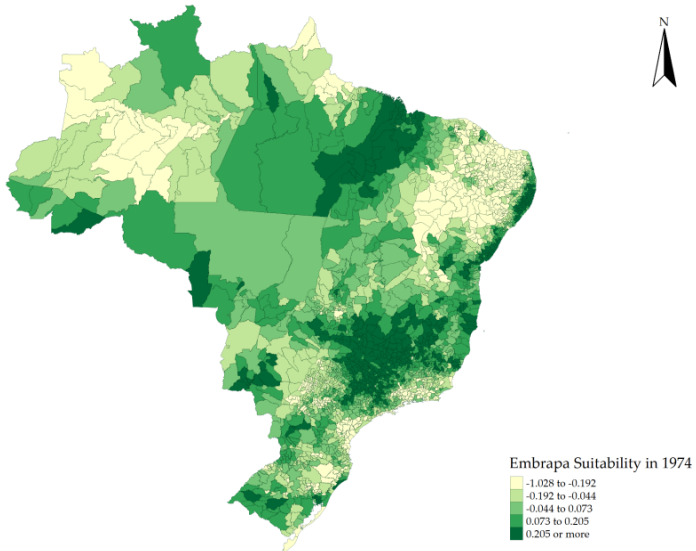
## (a) Similarity to Embrapa Trigo



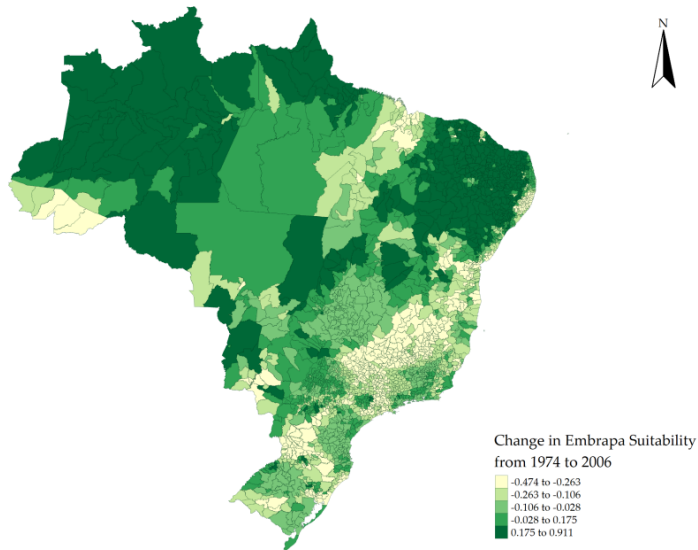
## (b) Similarity to Embrapa Pantanal



## (c) Embrapa Exposure in 1974



## (d) Change in Exposure, 1974 to 2006



## The comparison

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- When a center opens, compare municipalities ecologically **similar** to it with ecologically distant ones.
- Physical proximity is controlled for, and the result survives dropping municipalities within 100 km of any center.
- Assumption: similar and distant municipalities were on the same trend before the center opened.
- Placebo expansions of Embrapa across time and space put the real estimate in the right tail.

## Result: higher productivity

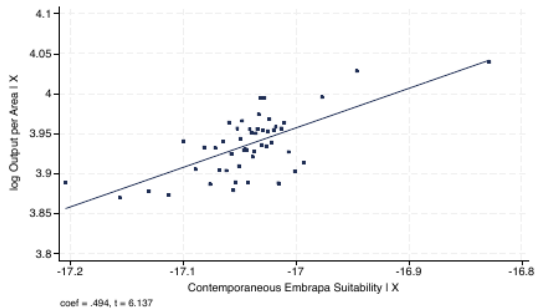
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- One standard deviation more exposure raised production value per unit of farm area by about **12 percent**.
- The same holds for land values, for crop yields, and for a broader productivity measure net of labor and inputs.
- No pre-trend over 1960 to 1970.
- The effects accumulate: the long-run effect is 30 to 40 percent larger than the within-decade effect.

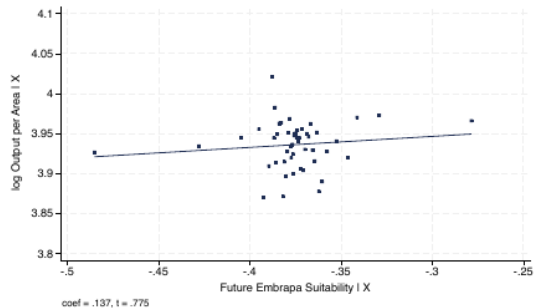


**Figure 7: Embrapa Affects Productivity Contemporaneously, with no Anticipation**

**(a) Contemporaneous Exposure**

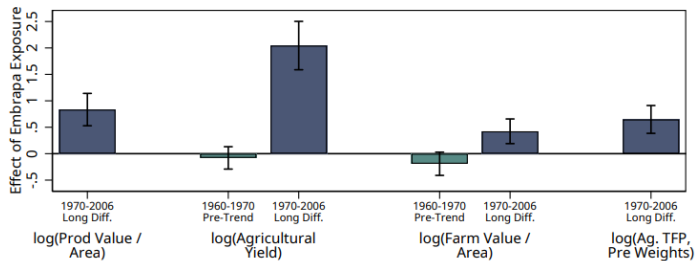


**(b) Future Exposure**

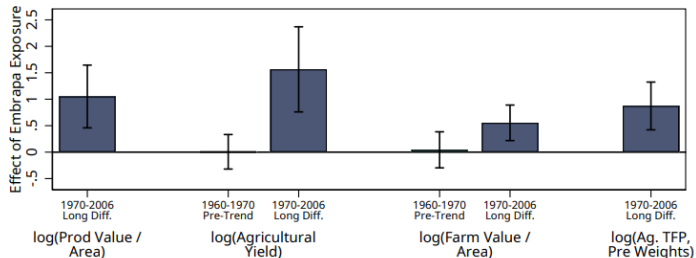


**Figure 8: Long Difference Estimates and Pre-Existing Trends**

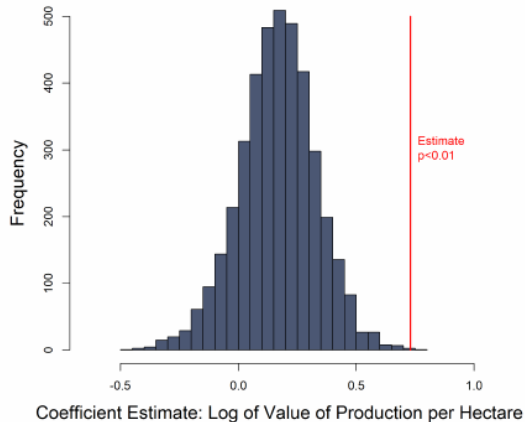
**(a) Unweighted**



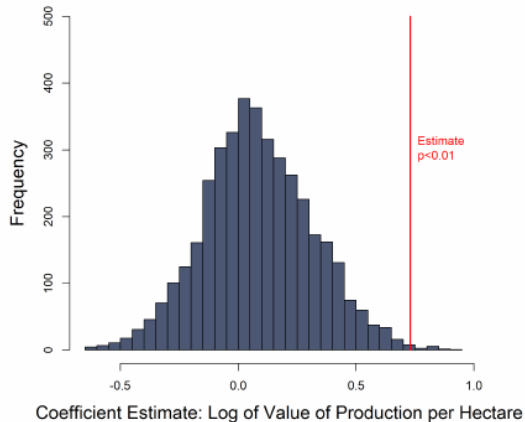
**(b) Weighted by Agricultural Area**



**Figure 9: Falsification Tests**



**(a)** Randomizing Across Geography and Time



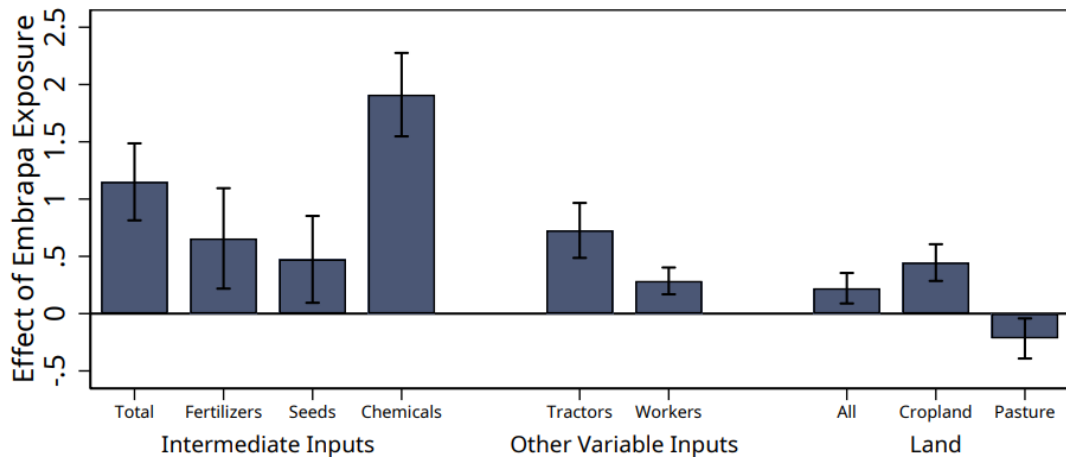
**(b)** Randomizing Across Time Only

# Mechanisms

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- Spending rises on exactly the inputs Embrapa worked on: fertilizers, seeds and chemical defenses.
- Cropland expands, over half of it converted from pasture. Labor use rises far less than input use.
- Embrapa's own **seed varieties** spread to ecologically similar municipalities even when far away: one standard deviation more similarity, about 50 percent more area planted.
- Output and yields rise for the focus crops and are flat for the rest, even within the same municipality and year.

**Figure 10: Embrapa Exposure and Agricultural Inputs**



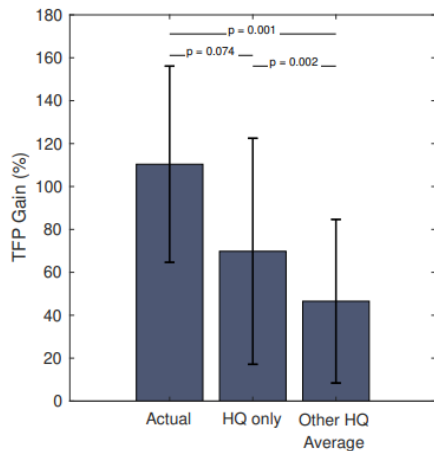
## Benefits and costs

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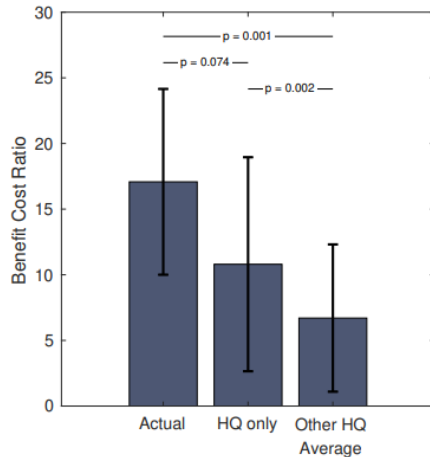
- Combining the estimates with a model: Embrapa raised aggregate agricultural productivity by **110 percent**.
- That is 39 percent of all Brazilian agricultural productivity growth since 1970.
- Benefit-cost ratio of 17, against a peak cost of about 1 percent of Brazil's agricultural GDP.
- For comparison, Jones and Summers put overall US public R&D at 5, and Griliches put hybrid corn at 7.

**Figure 11: Productivity Gains and Benefit-Cost Analysis of Embrapa**

**(a) Productivity Gains**



**(b) Benefit-Cost Ratio**



## The value of spreading out

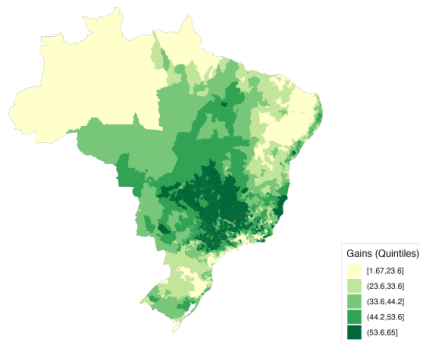
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- Counterfactual: the same budget concentrated in a single center in Brasilia.
- The productivity gain falls from 110 to 70 percent, and the benefit-cost ratio from 17 to 11.
- A single center at the average alternative location achieves less than half the observed benefit.
- Brasilia is ecologically close to much of the country, so it is the best single site, which is consistent with the choice made.
- **Geographic scope** carried much of the return, over and above the level of spending.

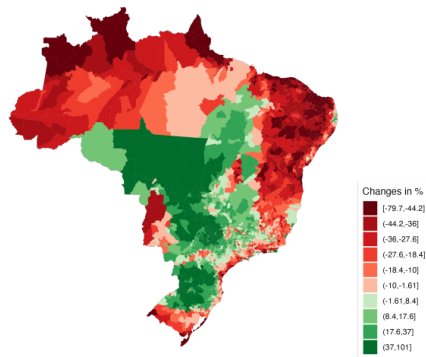


**Figure 12:** Geographic Distribution of the Productivity Gains of an Embrapa Center

**(a)** Initial Aggregate Productivity Gain



**(b)** Change in Aggregate Productivity Gain



## In short: Akerman and co-authors

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- Public R&D aimed at local conditions raised Brazilian agricultural productivity substantially.
- The mechanism was developing and diffusing technology matched to each region's ecology.
- The return was high by the standards of the literature on public R&D.
- Where the research capacity was placed did much of the work.

## Across the two papers

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- Aneja and Xu: shielding rank and file bureaucrats from politics improved what the state delivered.
- Akerman and co-authors: where the state placed its research capacity determined which technology got made.
- Both measure the state through an **output**: mail delivered correctly, crop yields.

# References

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- Aneja, A. and Xu, G. (2024). Strengthening State Capacity: Civil Service Reform and Public Sector Performance during the Gilded Age. *American Economic Review*, 114(8).
- Akerman, A., Moscona, J., Pellegrina, H. S. and Sastry, K. A. (2025). Public R&D Meets Economic Development: Embrapa and Brazil's Agricultural Revolution. Working paper.